Configuration Details:

System name of the submission: **MLFiles**
Submitter name: **Nutanix with Supermicro**
Submission date: **07/10/25**
Version of the benchmark: **2.0.0b1**
Solution type of the submission: **Shared File**
Submission division (OPEN or CLOSED): **CLOSED**
Power Requirements: **32500 Watts**

Host - Hypervisor

- AHV version 10.3
- 2x32 CPUs, Intel(R) Xeon(R) Gold 6448H @ 2.8 GHz
- 1024 GB DDR5 physical memory

CVM

- 32 vcpus
- 256GB RAM
- AOS version 7.3
- iSER data path enabled
- Compression disabled

FSVM

- 32 vcpus
- 200GB RAM (for Cosmoflow used 350GB)
- Files version 5.3
- SRIOV NIC for client traffic
- NFS over RDMA enabled
- RDMA max connection count 2048
- 2 FSVMs per host
- Performance profile
- Share compression disabled

Share and Data layout

- /root used general share
- Distributed shares submounted under /root:
    - Unet3d: /root/unet3d/train
    - Resnet50: /root/resnet50/train
    - Cosmoflow: /root/cosmoflow/train

- Top level directories:
    - Resnet50 and unet3d - 960 top level dirs
    - Cosmoflow - 1920 top level dirs
- 17 vdisks per FSVM for share
- Share workload type:
    - Cosmoflow - default share type
    - All other tests used sequential share type.

Clients

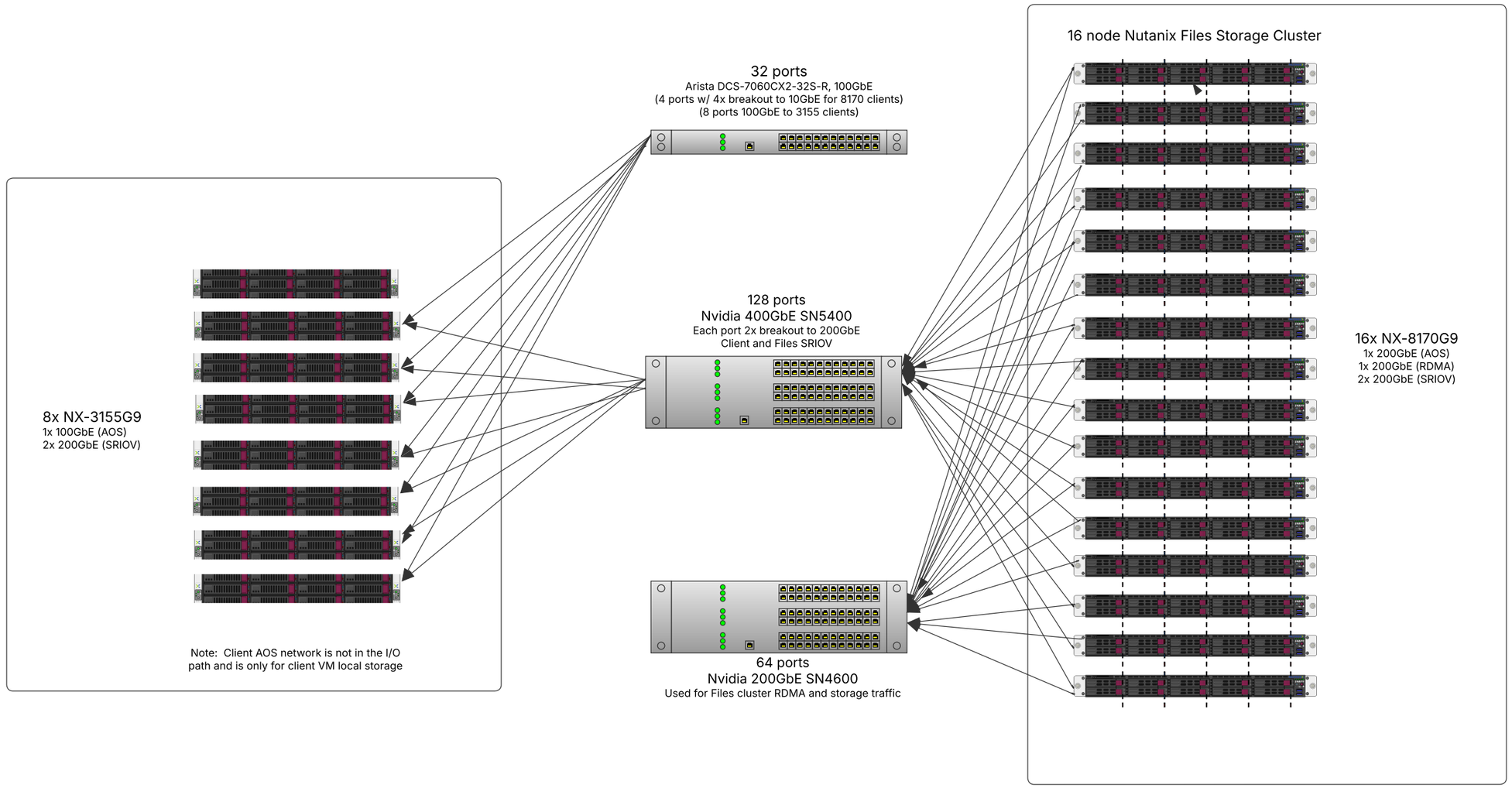
- Hosts - 8x NX-8170G8, 13x NX-3155G9 (note, some tests did not utilize all client hosts)
- VMs hosted on AHV 10.3, AOS 7.3 cluster
- 4x client VMs per host for G9 systems, 2x per host for G8.
- Ubuntu 24.04, kernel 6.9
- 8170-G8 hardware - SuperMicro, Intel Xeon Platinum 8352Y 2x32 CPUs
- 3155-G9 hardware - SuperMicro, Intel Xeon Gold 6548Y+, 2x32 CPUs
- 1024GB Physical RAM per host
- Client VMs
    - 32 vcpus
    - 64GB for cosmoflow
    - 192GB for resnet50
    - 192GB for unet3d
- Swap off
- NFS mount with RDMA, v4.0 and nconnect=4
- sunrpc.tcp_slot_table_entries=32
- NFS read_ahead - defaults
- Direct IO used for unet3d.

Example Benchmark Run Commands:

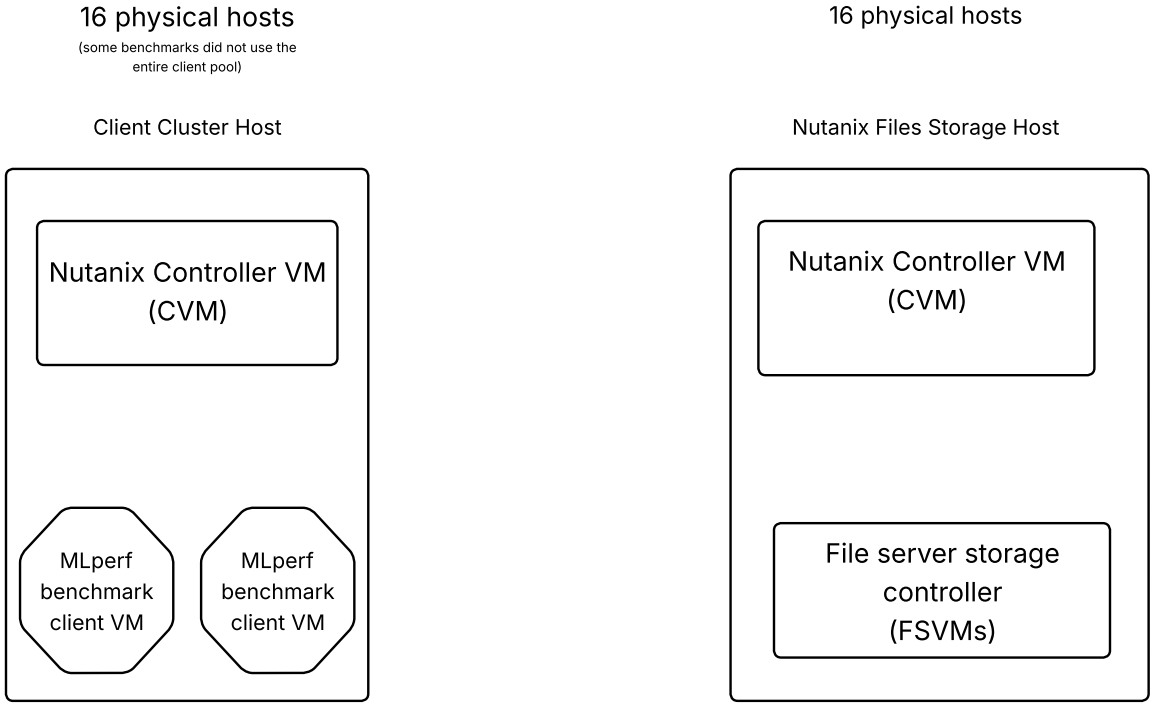
```
 mlpstorage training run --hosts
10.57.205.101,10.57.205.102,10.57.205.103,10.57.205.104,10.57.205.105,10.57.2
05.106,10.57.205.107,10.57.205.108,10.57.205.109,10.57.205.110,10.57.205.111,
10.57.205.112,10.57.205.113,10.57.205.114,10.57.205.115,10.57.205.116,10.57.2
05.117,10.57.205.118,10.57.205.119,10.57.205.120,10.57.205.121,10.57.205.122,
10.57.205.123,10.57.205.124,10.57.205.125,10.57.205.126,10. 57.205.127,10.57.2
05.128,10.57.205.129,10.57.205.130,10.57.205.131,10.57.205.132,10.57.205.133,
10.57.205.134,10.57.205.135,10.57.205.136,10.57.205.137,10.57.205.138,10.57.2
05.139,10.57.205.140,10.57.205.141,10.57.205.142,10.57.205.143,10.57.205.144,
10.57.205.145,10.57.205.146,10.57.205.147,10.57.205.148,10.57.205.149,10.57.2
05.150,10.57.205.151,10.57.205.152 --num-client-hosts 52
--client-host-mem ory-in-gb 64 --num-accelerators 52  --accelerator-type h100
--model unet3d --param dataset.num_files_train=20020 --data-dir /mnt/data
--results-dir unet3d_trial --allow-run-as-root --params
dataset.num_subfolders_train=960 reader.odirect=true --params
reader.read_threads=16
```

Example NFS mount command from Ubuntu client:

```
sudo mount -t nfs -o rdma -o vers=4.0 10.57.205.61:/resnet50 /mnt/resnet50
```


16 node Nutanix Files Storage Cluster
32 ports
Arista DCS-7060CX2-32S-R, 100GbE
(4 ports w/ 4x breakout to 10GbE for 8170 clients)
(8 ports 100GbE to 3155 clients)
128 ports
Nvidia 400GbE SN5400
Each port 2x breakout to 200GbE
Client and Files SRIOV
16x NX-8170G9
1x 200GbE (AOS)
1x 200GbE (RDMA)
2x 200GbE (SRIOV)
8x NX-3155G9
1x 100GbE (AOS)
2x 200GbE (SRIOV)
Note: Client AOS network is not in the I/O
path and is only for client VM local storage
64 ports
Nvidia 200GbE SN4600
Used for Files cluster RDMA and storage traffic

Virtual environment for each host:



System Compon	ePower Supply Unit	Nameplate Rated Power(W)	Design Power(W) *	Rack Unit
SN4600 switch	Power Supply 1	600	600	2
	Power Supply 2	600		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1

	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1
	Power Supply 2	2000		
8170G9	Power Supply 1	2000	2000	1

	Power Supply 2	2000		

Total		65200	32600	18

*Note: All power supplies are redundant, so Design power is half of Nameplate power.